# Green Geometry of Positive Hermitian Instruments: Hyperbolic Transport, Relational Holonomy, and Fixed Five-Functional Tomography

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**Abstract**

We study the cone $\mathcal{P}$ of positive Hermitian $2\times 2$ matrices with the Green/Riesz metric $g_G(U,V) = \mathrm{Re}\,\mathrm{tr}(G^{-1}UG^{-1}V)$. A direct calculation gives its Levi–Civita connection and splits the geometry into a flat scale direction and determinant-one shape leaves of sectional curvature $-1/2$. Parallel transport around a closed instrument loop acts on the traceless Hermitian directions by a rotation in SO(3). This return is relational: it is visible against a fixed reference frame and disappears when the reference is co-transported. We then ask how few real-linear readings determine the return rotation. Three antisymmetric readings give local coordinates, while adding the trace still fails at half-turns. Five fixed functionals suffice globally. The proof combines six rank-two projections in $\mathbb{R}^4$, exact whitening, the quaternion–rotation dictionary, and an exact real-kernel exclusion. By the published identity $m_{\mathbb{R}}(4) = 6$, five is also minimal among fixed real-linear schemes. For the resulting positive-projector design we prove a kernel floor $1/1024$, a two-sided rotation-angle estimate, and condition number at most 560.681. A separate rational four-plane has proved floor $1/32$ and an explicit abstract isometric realization with a larger lower angle constant. These are comparisons of proved bounds, not of unknown true optima; no stochastic-noise, device, or physical-realization conclusion is drawn.

## 1 Introduction

Positive-definite matrix geometry is classical. In particular, the affine-invariant trace metric, its congruence invariance, geodesics, connection, and nonpositive curvature belong to the established geometry of positive matrices [20, 19, 8, 25, 15, 9]. Its covariance Fisher–Rao geometry is likewise established [24, 1, 18]. Quantum-state monotone metrics are a distinct comparison family [21]; no quantum statistical interpretation is inferred here. The determinant-one positive Hermitian $2 \times 2$ model is the standard hyperbolic $\mathrm{SL}(2,\mathbb{C})/\,\mathrm{SU}(2)$ model [10].

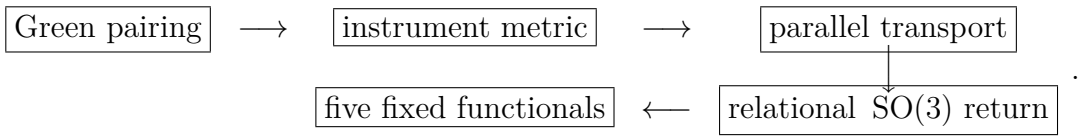
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Likewise, attitude estimation, rotation synchronization, continuous Euclidean representations of SO(3), and real generalized phase retrieval are established subjects [26, 27, 31, 30, 28].

The return matrix $C_{ab} = \frac{1}{2}\operatorname{tr}(\sigma_a H \sigma_b H^\dagger)$ is also the Pauli transfer matrix of a unitary qubit channel. The closest tomographic comparison is therefore process tomography under a unitarity promise, rather than state tomography [13, 4, 11]. Our count concerns exact synthesized real-linear coordinates on SO(3); it is not a count of experimental settings, samples, or interactive observables in those works.

Our question has two parts. What does an observer who retains a fixed Green reference frame see after parallel transport around a loop? And how few real-linear readings determine that return? We derive the geometry from the Green/Riesz pairing, express the return as a relational rotation, and then solve the associated fixed-functional reconstruction problem.

The mathematical chain is




Every arrow is finite-dimensional and exact. The geometry is mathematical: we do not infer an evolution law or physical realization of arbitrary loops.

The paper has five main steps. First, Theorem 2.4 derives the metric connection, curvature, and scale–shape product. Second, Proposition 3.1 and Theorem 4.2 derive exact transport and relational holonomy. Third, Theorem 5.2 and Proposition 5.3 distinguish local coordinates from a global wall. Fourth, Theorems 7.2 and 7.3 give a fixed, nonadaptive, globally injective five-functional construction. Corollary 10.1 uses the known phase-retrieval count to obtain fixed-linear optimality, and Proposition 11.2 records the adaptive comparison. Theorem 8.1 and Corollary 8.2 give quantitative bounds for the positive-projector construction. Finally, Theorem 9.1 and Corollary 9.2 record a separately specified rational four-plane with a larger proved conditioning floor. The two designs serve different purposes: the first is tied to positive projections, whereas the second is an abstract measurement kernel.

## 2 The Green metric and hyperbolic shape geometry

**Definition 2.1** (Instrument and tangent carrier)**.** Let

$$\mathcal{P} = \{G \in \operatorname{Herm}(2) : G > 0\}.$$

The canonical tangent identification is $T_G\mathcal{P} = \operatorname{Herm}(2)$. Under an invertible passive complex frame $W$, instrument and tangent transform by the same congruence,

$$G' = W^{-\dagger} G W^{-1}, \qquad U' = W^{-\dagger} U W^{-1}. \tag{1}$$

**Definition 2.2** (Green source and reading)**.** For $G \in \mathcal{P}$ and $Q \in \operatorname{Herm}(2)$, the Green/Riesz source is $S_Q = G^{-1}Q$. The scalar reading of a tangent response $U$ is

$$\mathcal{E}_G(Q, U) = \operatorname{Re}\operatorname{tr}(G^{-1}QG^{-1}U). \tag{2}$$

We apply the same pairing to tangent variations and write

$$g_G(U, V) = \operatorname{Re}\operatorname{tr}(G^{-1}UG^{-1}V). \tag{3}$$

**Lemma 2.3** (Determinant tangent and scale split)**.** *For $G \in \mathcal{P}$,*

$$d(\det)_G[U] = \det(G)\,\mathrm{tr}(G^{-1}U). \tag{4}$$

*Consequently the tangent space to a fixed-determinant leaf is*

$$T_G\mathcal{P}_{\det G} = \{U \in \mathrm{Herm}(2) : \mathrm{tr}(G^{-1}U) = 0\},$$

*and the exact projection onto it is*

$$\Pi_G(U) = U - \tfrac{1}{2}\,\mathrm{tr}(G^{-1}U)G. \tag{5}$$

*The scale vector $G$ is orthogonal to this leaf and $g_G(G, G) = 2$.*

*Proof.* Jacobi's formula gives (4). Since $\mathrm{tr}(G^{-1}G) = 2$, equation (5) is trace-free in the required sense. Finally, $g_G(G, U) = \mathrm{tr}(G^{-1}U)$ and $g_G(G, G) = \mathrm{tr}\, I_2 = 2$. □

The next theorem derives the normalization of the classical affine-invariant positive-matrix geometry induced by this Green/Riesz pairing [20, 19, 8, 25, 15].

**Theorem 2.4** (Green geometry)**.** *The pairing (3) is a positive Riemannian metric, invariant under (1). Its unique torsion-free, metric-compatible connection is*

$$\nabla_X Y = DY[X] - \tfrac{1}{2}\Big(XG^{-1}Y + YG^{-1}X\Big). \tag{6}$$

*With the curvature convention $R(X, Y)Z = \nabla_X\nabla_Y Z - \nabla_Y\nabla_X Z - \nabla_{[X,Y]}Z$,*

$$\begin{aligned} R_G(X, Y)Z = -\tfrac{1}{4}\Big(&XG^{-1}YG^{-1}Z - YG^{-1}XG^{-1}Z \\ &- ZG^{-1}XG^{-1}Y + ZG^{-1}YG^{-1}X\Big). \end{aligned} \tag{7}$$

*The fixed-determinant leaves are totally geodesic. Every nondegenerate two-plane tangent to such a leaf has sectional curvature*

$$K = -\tfrac{1}{2}. \tag{8}$$

*If*

$$G = e^s P, \qquad \det P = 1, \tag{9}$$

*then the metric splits exactly as*

$$g_G = 2\,ds^2 + g_P. \tag{10}$$

*Thus the full cone is nonpositively curved but not constant-curvature: scale–shape planes are flat.*

*Proof.* Put $A = G^{-1}$ and $H = A^{1/2}UA^{1/2}$. Then $H$ is Hermitian and

$$g_G(U, U) = \mathrm{tr}(H^2) > 0$$

for $U \neq 0$. Under (1), $(G')^{-1} = WG^{-1}W^\dagger$; cyclicity of trace proves invariance.

To derive, rather than import, the connection, use $D_X A = -AXA$. For constant Hermitian vector fields,

$$D_X g(Y, Z) = -\,\mathrm{Re}\,\mathrm{tr}(AXAYAZ + AYAXAZ).$$

Insert this identity and its cyclic permutations into the Koszul formula. For the general symmetric one-coefficient bilinear ansatz $\Gamma_G(X,Y) = \alpha(XAY + YAX)$, cyclic trace reduction leaves $2\alpha + 1 = 0$. Hence $\alpha = -1/2$, which gives (6). Symmetry in $X, Y$ gives zero torsion; substituting $D_X A = -AXA$ gives $\nabla g = 0$. Uniqueness is the standard uniqueness part of the Koszul construction.

Differentiate (6), again using $D_X A = -AXA$, and subtract the expression with $X, Y$ exchanged. The derivative terms cancel and leave (7). Whitening by $G^{-1/2}$ reduces it to

$$R(X,Y)Z = -\tfrac{1}{4}[[X,Y],Z], \tag{11}$$

and therefore

$$\langle R(X,Y)Y, X\rangle = \tfrac{1}{4}\operatorname{tr}([X,Y]^2) \le 0. \tag{12}$$

For traceless Hermitian $2 \times 2$ matrices,

$$-\operatorname{tr}([X,Y]^2) = 2\{\operatorname{tr}(X^2)\operatorname{tr}(Y^2) - \operatorname{tr}(XY)^2\}.$$

Division by the Gram determinant in (12) gives $K = -1/2$.

If tangent fields $X, Y$ obey $\operatorname{tr}(AX) = \operatorname{tr}(AY) = 0$, differentiating the first constraint in the $Y$ direction and using (6) shows $\operatorname{tr}(A\nabla_Y X) = 0$. Hence the leaf is totally geodesic. Finally, $G^{-1}dG = ds\, I + P^{-1}dP$ in the trace calculation, and $\operatorname{tr}(P^{-1}dP) = d\log\det P = 0$. The cross term vanishes and $\operatorname{tr} I_2 = 2$, giving (10). □

*Remark* 2.5 (Classical identification). Theorem 2.4 recovers the classical affine-invariant positive-matrix metric [25, 15]. On $\det P = 1$, the conventional hyperbolic metric has curvature $-1$; the normalization (3) is twice that metric and therefore has curvature $-1/2$ [10]. No new affine-invariant metric or hyperbolic model is claimed. Geometrically, radial motion changes only the overall scale, while motion inside a determinant-one leaf follows the three-dimensional hyperbolic shape geometry. Curvature then predicts the leading small-loop return: an infinitesimal loop of oriented area $A$ in an orthonormal shape plane produces a rotation of order $A/2$.

# 3 Exact geodesics and parallel transport

Affine-invariant positive-matrix geodesics are classical [9]; this section derives the endpoint formula used by the Green normalization above.

The geodesic equation obtained by setting $\nabla_{\dot G}\dot G = 0$ in (6) is

$$\ddot G = \dot G\, G^{-1}\dot G. \tag{13}$$

The parallel-field equation along the same path is

$$\dot U = \tfrac{1}{2}\big(\dot G\, G^{-1}U + UG^{-1}\dot G\big). \tag{14}$$

**Proposition 3.1** (Exact endpoint transport)**.** *Let $G, H \in \mathcal{P}$ satisfy $\det G = \det H$. Put*

$$M = HG^{-1}, \qquad E_{G\to H} = \frac{M + I}{\sqrt{\operatorname{tr}(M) + 2}},$$

*where the square root in the denominator is the positive one. Then*

$$E_{G\to H}^2 = M, \quad E_{G\to H}G = GE_{G\to H}^\dagger, \quad E_{G\to H}GE_{G\to H}^\dagger = H, \quad \det E_{G\to H} = 1. \tag{15}$$

*Parallel transport along the affine-invariant geodesic from $G$ to $H$ is*

$$\mathcal{T}_{G\to H}(U) = E_{G\to H}UE_{G\to H}^\dagger. \tag{16}$$

*It preserves Hermiticity, the determinant-tangent constraint, and the metric, and its reverse is $\mathcal{T}_{H\to G}$. Segment transports compose in chronological order. Under the passive congruence (1), the transport is covariant:*

$$\mathcal{T}_{G'\to H'}(U') = W^{-\dagger}\mathcal{T}_{G\to H}(U)W^{-1}. \tag{17}$$

*Equal determinants are needed for the displayed closed form and for the path to stay on one determinant leaf. For arbitrary positive $G, H$, the same transport formula holds with $E_{G\to H} = (HG^{-1})^{1/2}$, using the positive spectral square root.*

*Proof.* $M$ is similar to the positive Hermitian matrix $G^{-1/2}HG^{-1/2}$, so its eigenvalues are positive. Equal determinants give $\det M = 1$. Cayley–Hamilton then yields $M^2 - (\operatorname{tr} M)M + I = 0$, hence $(M+I)^2 = (\operatorname{tr} M + 2)M$ and the first identity in (15). Since $MG = H = GM^\dagger$, every real polynomial in $M$ satisfies the analogous $G$-adjoint identity; this proves the second identity. The third follows from $EGE^\dagger = E^2G = MG = H$, and the positive square root has determinant one.

Define $G(t) = M^tG$ by the positive spectral calculus. Direct differentiation gives (13). Substitution shows that $U(t) = M^{t/2}U(0)(M^\dagger)^{t/2}$ solves (14); at $t = 1$ it is (16). The remaining assertions follow by congruence, cyclicity of trace, composition of the segment maps, and reversal of the geodesic. In particular, $M' = W^{-\dagger}MW^\dagger$ and $E_{G'\to H'} = W^{-\dagger}E_{G\to H}W^\dagger$, which gives (17). □

# 4 Closed loops and relational Green holonomy

Holonomy of symmetric spaces is classical [16, 2]. Berry-phase anholonomy is a useful neighboring comparison [7], but the return below is a relational Green-frame rotation; it is not a physical quantum phase or an adiabatic-time claim.

**Definition 4.1** (Relational return)**.** Let $G_0, \dots, G_n$ be positive determinant-one instruments with $G_n = G_0 = I$. In chronological order put

$$H_{\mathcal{L}} = E_{G_{n-1}\to G_n} \cdots E_{G_1\to G_2}E_{G_0\to G_1}. \tag{18}$$

For the Pauli frame $(\sigma_1, \sigma_2, \sigma_3)$, define

$$C_{ab}(\mathcal{L}) = \tfrac{1}{2}\operatorname{tr}\left(\sigma_a H_{\mathcal{L}}\sigma_b H_{\mathcal{L}}^\dagger\right). \tag{19}$$

Because (18) returns the base instrument, $H_{\mathcal{L}}H_{\mathcal{L}}^\dagger = I$ and $\det H_{\mathcal{L}} = 1$. Thus $H_{\mathcal{L}} \in \mathrm{SU}(2)$ and $C(\mathcal{L}) \in \mathrm{SO}(3)$. The rotation-matrix readout lies within the classical literature on attitude estimation and rotation synchronization [26, 27]; the statement here concerns its exact derivation from retained versus transported Green reference frames.

**Theorem 4.2** (Relational holonomy and source readout)**.** *For every determinant-one loop based at $I$,*

$$C_{ab}(\mathcal{L}) = \tfrac{1}{2} g_I\big(\sigma_a, H_{\mathcal{L}} \sigma_b H_{\mathcal{L}}^\dagger\big)\,. \tag{20}$$

*If both the query and reference frames are co-transported, their mutual reading is instead*

$$\tfrac{1}{2} g_I(H_{\mathcal{L}} \sigma_a H_{\mathcal{L}}^\dagger, H_{\mathcal{L}} \sigma_b H_{\mathcal{L}}^\dagger) = \delta_{ab}. \tag{21}$$

*Loop reversal gives $C^{-1}$, concatenation gives the ordered product, and $H_{\mathcal{L}}$ and $-H_{\mathcal{L}}$ have the same readout. Noncollinear loops can have $C \neq I$, while collinear geodesic loops return $I$. The infinitesimal return algebra on the three Pauli modalities is all of $\mathfrak{so}(3)$.*

*Proof.* At $G = I$, equation (3) is the Hilbert–Schmidt pairing, so (20) is exactly (19). Metric preservation proves (21). Functoriality of parallel transport gives the reversal and concatenation laws, and the adjoint action is blind to the center of SU(2).

For traceless Hermitian $X, Y$, the small-loop return generator is $[X, Y]/4$ with the boundary orientation fixed above. Its adjoint action on the Pauli frame is a real skew $3 \times 3$ matrix. The three independent Pauli pairs produce three independent generators and close under commutator as $\mathfrak{so}(3)$. This is the local holonomy algebra, consistent with the classical curvature–holonomy relation [2]. Exact finite nonidentity and collinear examples are displayed below. □

Here are three exact loops realizing nontrivial returns. For $a, b > 0$ and $|z| = 1$, write

$$\mathcal{D}(a) = \begin{pmatrix} a^2 & 0 \\ 0 & a^{-2} \end{pmatrix}, \qquad \mathcal{B}(b, z) = \begin{pmatrix} c_b & z s_b \\ \bar{z} s_b & c_b \end{pmatrix}, \quad c_b = \frac{b^2 + b^{-2}}{2}, \quad s_b = \frac{b^2 - b^{-2}}{2}.$$

Both matrices are positive Hermitian with determinant one.

**Example 4.3** (A real-axis triangle)**.** The geodesic triangle $I \to \mathcal{D}(7/3) \to \mathcal{B}(5/3, 1) \to I$ has return

$$C_1 = \begin{pmatrix} 217449/268649 & 0 & -157760/268649 \\ 0 & 1 & 0 \\ 157760/268649 & 0 & 217449/268649 \end{pmatrix}. \tag{22}$$

It fixes the second Pauli axis.

**Example 4.4** (An imaginary-axis triangle)**.** The triangle $I \to \mathcal{D}(8/5) \to \mathcal{B}(9/4, \mathrm{i}) \to I$ has return

$$C_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 34051232/40477457 & 21884655/40477457 \\ 0 & -21884655/40477457 & 34051232/40477457 \end{pmatrix}. \tag{23}$$

It fixes the first Pauli axis.

**Example 4.5** (A generic complex-axis triangle)**.** For $z = (3 + 4\mathrm{i})/5$, the triangle $I \to \mathcal{D}(11/7) \to \mathcal{B}(6/5, z) \to I$ has return

$$C_3 = \begin{pmatrix} 673203337/676026025 & 3763584/676026025 & -2463912/27041041 \\ 3763584/676026025 & 671007913/676026025 & 3285216/27041041 \\ 2463912/27041041 & -3285216/27041041 & 26727409/27041041 \end{pmatrix}. \tag{24}$$

Its fixed axis is not a coordinate Pauli axis. By contrast, every collinear loop $I \to \mathcal{D}(a) \to \mathcal{D}(b) \to I$ has identity return.

# 5 Local tomography and the half-turn wall

Minimal tomography for trace-one qubits and informational completeness under prior information are established, with measurement objects and counts that differ from the unnormalized positive-cone Green-response model used here [22, 14]. Consequently, every minimality statement below is restricted to its declared reading scheme or to the separately stated class of fixed real-linear functionals; it is not a general tomography lower bound.

**Definition 5.1** (Antisymmetric local reading)**.** For $C \in \mathrm{SO}(3)$, set

$$w(C) = \tfrac{1}{2}\begin{pmatrix} C_{32} - C_{23} \\ C_{13} - C_{31} \\ C_{21} - C_{12} \end{pmatrix}. \tag{25}$$

For $w \in \mathbb{R}^3$, let $[w]_\times x = w \times x$.

**Theorem 5.2** (Three local readings)**.** *The differential of* (25) *at the identity has rank three. On the branch*

$$c = \tfrac{1}{2}(\operatorname{tr} C - 1) > 0,$$

*the three readings determine $C$ exactly by*

$$c = +\sqrt{1 - w^{\mathsf{T}} w}, \qquad C = cI + [w]_\times + \frac{ww^{\mathsf{T}}}{1+c}. \tag{26}$$

*No continuous scalar instrument with fewer than three outputs is locally injective on an open subset of the three-dimensional return manifold.*

*Proof.* Write a rotation near the identity as $C = \exp([\omega]_\times)$. Then $w(C) = \omega + O(|\omega|^2)$, proving the rank statement. Rodrigues' formula gives $w = \sin\theta\, n$ and $c = \cos\theta$; on $c > 0$ the positive square-root choice is forced and rearrangement yields (26). Exact multiplication verifies $C^{\mathsf{T}}C = I$, $\det C = 1$, and readback. The lower bound is invariance of dimension: an open subset of a three-manifold cannot embed continuously into $\mathbb{R}^m$ for $m < 3$. □

**Proposition 5.3** (Natural four-reading wall)**.** *The four readings $(w(C), \operatorname{tr} C)$ are not globally injective. Every half-turn about a unit axis $n$ has*

$$C_n = 2nn^{\mathsf{T}} - I, \qquad (w(C_n), \operatorname{tr} C_n) = (0, 0, 0, -1). \tag{27}$$

*Proof.* $C_n$ is symmetric, so its antisymmetric reading is zero. Its eigenvalues are $(1, -1, -1)$, so its trace is $-1$. Distinct axes not related by sign give distinct rotations with the same four-reading code. □

# 6 Five fixed real-linear Green functionals

**Definition 6.1** (Fixed functional)**.** A fixed real-linear Green functional is

$$\ell_K(C) = \langle K, C \rangle_F = \operatorname{tr}(K^{\mathsf{T}}C), \qquad K \in \mathbb{R}^{3\times 3},$$

where $K$ is selected before $C$ is known. Because the nine entries of $C$ are the fixed-reference readings (20), $\ell_K$ lies in their real-linear span. Fixed excludes data-dependent pivots or a change of $K$ after observing $C$.

Use quaternion order $q = (w, x, y, z)^{\mathsf{T}}$ and the standard double cover $q \mapsto R(q)$:

$$R(q) = \begin{pmatrix} w^2 + x^2 - y^2 - z^2 & 2(xy - wz) & 2(xz + wy) \\ 2(xy + wz) & w^2 - x^2 + y^2 - z^2 & 2(yz - wx) \\ 2(xz - wy) & 2(yz + wx) & w^2 - x^2 - y^2 + z^2 \end{pmatrix}. \tag{28}$$

For unit $q$, $R(q) \in \mathrm{SO}(3)$ and $R(q) = R(-q)$.

**Lemma 6.2** (Quaternion–rotation dictionary)**.** *Let $D = (d_{\mu\nu})_{\mu,\nu=0}^{3}$ be real symmetric and trace-free. Define $K(D) \in \mathbb{R}^{3\times 3}$ by*

$$\begin{aligned} K_{11} &= (d_{00} + d_{11})/2, & K_{22} &= (d_{00} + d_{22})/2, & K_{33} &= (d_{00} + d_{33})/2, \\ K_{23} &= (d_{23} - d_{01})/2, & K_{32} &= (d_{23} + d_{01})/2, & & \\ K_{13} &= (d_{02} + d_{13})/2, & K_{31} &= (d_{13} - d_{02})/2, & & \\ K_{12} &= (d_{12} - d_{03})/2, & K_{21} &= (d_{12} + d_{03})/2. & & \end{aligned} \tag{29}$$

*Then, coefficientwise for every $q \in \mathbb{R}^4$,*

$$q^{\mathsf{T}} D q = \langle K(D), R(q) \rangle_F \,. \tag{30}$$

*Together with $q^{\mathsf{T}} q$, the nine quadratic entries of $R(q)$ span* Sym(4) *with rank ten.*

*Proof.* Substitute (28) into (30). Direct coefficient comparison gives $d_{00}, d_{11}, d_{22}, d_{33}$ on the diagonal, using $\operatorname{tr} D = 0$, and $2d_{\mu\nu}$ on each cross term. This proves the identity. The inverse recovers all nine trace-free coordinates, while the norm recovers the scalar direction, giving rank ten. □

The fixed construction starts from Xu's six rank-two projections [30, Theorem 4.2], represented here by the columns $u_j, v_j$ of

$$U = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & -4 & 2 \\ 0 & 0 & 0 & 1 & -3 & 3 \end{pmatrix}, \qquad V = \begin{pmatrix} 0 & 5 & -1 & -1 & -17 & -5 \\ -5 & 0 & 0 & 5 & 4 & 4 \\ 2 & -2 & 0 & -3 & -2 & 2 \\ -2 & 1 & 0 & 0 & -3 & -1 \end{pmatrix}. \tag{31}$$

Define

$$P_j = \frac{u_j u_j^{\mathsf{T}}}{u_j^{\mathsf{T}} u_j} + \frac{v_j v_j^{\mathsf{T}}}{v_j^{\mathsf{T}} v_j}, \qquad S = \sum_{j=1}^{6} P_j. \tag{32}$$

The exact factorization used below is $S = LDL^{\mathsf{T}}$, with

$$L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -1206036/7353455 & 1 & 0 & 0 \\ -629974/7353455 & -620743491/3418884916 & 1 & 0 \\ 865837/7353455 & 562593553/3418884916 & 39888940419/121436249827 & 1 \end{pmatrix} \tag{33}$$

and

$$D = \operatorname{diag}\left( \frac{1470691}{332787}, \frac{3418884916}{1213320075}, \frac{121436249827}{51283273740}, \frac{213445555084}{121436249827} \right). \tag{34}$$

Put

$$C_{\mathrm{w}} = D^{-1/2} L^{-1}, \qquad B_j = C_{\mathrm{w}} P_j C_{\mathrm{w}}^{\mathsf{T}}. \tag{35}$$

**Lemma 6.3** (Exact projector and whitening identities)**.** *For every $j$, $u_j^{\mathsf{T}} v_j = 0$, $P_j = P_j^{\mathsf{T}} = P_j^2$, $P_j \geq 0$, and* $\operatorname{rank} P_j = 2$*. The four diagonal entries of (34) are positive, $S = LDL^{\mathsf{T}}$, and*

$$\sum_{j=1}^{6} B_j = I_4. \tag{36}$$

*Proof.* All claims before the factorization follow directly from the six integer column pairs in (31); orthogonality makes (32) the sum of two orthogonal rank-one projections. Every diagonal entry of $D$ has positive numerator and denominator. Rational multiplication verifies $S = LDL^{\mathsf{T}}$. Therefore

$$\sum_j B_j = D^{-1/2} L^{-1} (LDL^{\mathsf{T}}) L^{-\mathsf{T}} D^{-1/2} = I_4.$$

□

For each $j$, write

$$\alpha_j = \tfrac{1}{4} \operatorname{tr} B_j, \qquad D_j = B_j - \alpha_j I_4, \qquad K_j = K(D_j). \tag{37}$$

The five promised functionals are

$$\ell_j(C) = \langle K_j, C \rangle_F, \qquad j = 1, \ldots, 5. \tag{38}$$

Equations (31)–(37) are the exact definition of every coefficient; no decimal or fitted parameter enters.

# 7 Exact rank-two-kernel reconstruction and global injectivity

**Lemma 7.1** (Real phase-retrieval kernel criterion)**.** *Let $B_1, \ldots, B_m$ be real symmetric matrices with $\sum_j B_j = I$. The quadratic map*

$$q \longmapsto (q^{\mathsf{T}} B_1 q, \ldots, q^{\mathsf{T}} B_m q)$$

*determines every $q \in \mathbb{R}^4$ up to sign if and only if its symmetric kernel contains no nonzero matrix of rank at most two.*

*Proof.* If $q$ and $r$ have equal measurements, then $Q = qq^{\mathsf{T}} - rr^{\mathsf{T}}$ is a symmetric kernel element of rank at most two. Conversely, let a nonzero symmetric kernel element $Q$ have rank at most two. Since $\sum B_j = I$, $\operatorname{tr} Q = 0$. It cannot be nonzero semidefinite, so its spectral form is $\lambda aa^{\mathsf{T}} - \lambda bb^{\mathsf{T}}$ with $\lambda > 0$ and orthonormal $a, b$. The two non-antipodal vectors $\sqrt{\lambda}a, \sqrt{\lambda}b$ then have equal quadratic measurements. □

**Theorem 7.2** (Exact six-map kernel exclusion)**.** *For the six projectors (32), the linear map*

$$\mathcal{M}(Q) = (\operatorname{tr}(P_1 Q), \ldots, \operatorname{tr}(P_6 Q)), \qquad Q \in \operatorname{Sym}(4),$$

*has rank six and a four-dimensional kernel, and that kernel contains no nonzero real matrix of rank at most two. Consequently both $(P_j)_{j=1}^6$ and their congruence-whitened family $(B_j)_{j=1}^6$ perform real phase retrieval in dimension four.*

*Proof.* Use symmetric coordinate order

$$(q_{00}, q_{01}, q_{02}, q_{03}, q_{11}, q_{12}, q_{13}, q_{22}, q_{23}, q_{33}).$$

Exact row reduction of the $6 \times 10$ measurement matrix gives rank six and the four rational kernel vectors printed in the exact supplement. Write a general kernel matrix as

$$Q(a, b, c, d) = aQ_a + bQ_b + cQ_c + dQ_d.$$

A symmetric $4 \times 4$ matrix has rank at most two exactly when all ten upper-triangular entries of its adjugate (the corresponding $3 \times 3$ cofactors of the matrix) vanish. Homogeneity permits the projective cover

$$d = 1, \quad d = 0, c = 1, \quad d = c = 0, b = 1, \quad d = c = b = 0, a = 1.$$

On $d = 1$, exact lexicographic elimination gives a three-element basis and a degree-ten univariate eliminant in $c$. Its complete integer coefficients are printed in the supplement. A signed-remainder Sturm chain has degree sequence

$$10, 9, 8, 7, 6, 5, 4, 3, 2, 1, 0$$

and five sign variations at each of $-\infty$ and $+\infty$, hence no real root. The $d = 0, c = 1$ and $d = c = 0, b = 1$ ideals both contain 1. On the last patch, clear denominators from each cofactor polynomial, divide its integer coefficient content, and fix its sign. This nonzero rational rescaling preserves the common determinantal zero locus. The first declared nonzero primitive denominator-cleared cofactor polynomial then evaluates to

$$2324385247266000.$$

Thus the projective cofactor system has no real point, proving the kernel claim. Invertible congruence preserves rank and transports kernels between $P_j$ and $B_j$, so the whitened family has the same property. □

**Theorem 7.3** (Five fixed-function global tomography)**.** *The fixed map*

$$\Phi : \mathrm{SO}(3) \longrightarrow \mathbb{R}^5, \qquad \Phi(C) = (\ell_1(C), \ldots, \ell_5(C)), \tag{39}$$

*defined by* (31)–(38) *is injective. All five $K_j$ have exact rank three and nonzero determinant.*

*Proof.* Let $C = R(q)$ with $q^{\mathsf{T}} q = 1$. By Lemma 6.2,

$$y_j := q^{\mathsf{T}} B_j q = \alpha_j + \ell_j(C). \tag{40}$$

The five values in (39) therefore recover $y_1, \ldots, y_5$. Tightness (36) and unit norm give

$$y_6 = 1 - \sum_{j=1}^{5} y_j. \tag{41}$$

Theorem 7.2 says that the six $y_j$ determine $q$ up to the unavoidable sign $q \sim -q$. Since $R(q) = R(-q)$, they determine $C$ uniquely. Exact determinant evaluation gives the five nonzero values listed in the supplement, proving the rank statement. □

*Remark* 7.4 (What five counts). The five objects in (38) are synthesized real-linear functionals of the nine primitive matrix-entry readings. Each coefficient matrix has rank three. The sum of those ranks, 15, is only an algebraic upper bound on a rank-one coefficient decomposition. It is neither a primitive Green-shot count nor an optimal implementation cost.

## 7.1 Kernel floors and angle constants

Let $\mathcal{S}_0$ be the space of trace-zero real symmetric $4 \times 4$ matrices with Frobenius inner product, and let

$$\mathcal{W} = \{aa^\mathsf{T} - bb^\mathsf{T} : a, b \in \mathbb{R}^4,\ a^\mathsf{T}a = b^\mathsf{T}b = 1,\ a^\mathsf{T}b = 0\}. \tag{42}$$

Every $Z \in \mathcal{W}$ has $\|Z\|_F^2 = 2$. For a four-plane $K \subset \mathcal{S}_0$, with Frobenius projection $\Pi_K$, define its *kernel floor* by

$$m(K) = \min_{Z \in \mathcal{W}} \|\Pi_{K^\perp} Z\|_F^2 = 2 - \max_{Z \in \mathcal{W}} \|\Pi_K Z\|_F^2. \tag{43}$$

This is the normalization used for both four-planes below. If $K_1, \dots, K_4$ is a basis of $K$, put $\Gamma_{ij} = \operatorname{tr}(K_i K_j)$ and $t_K(Z)_i = \operatorname{tr}(K_i Z)$. Then

$$\|\Pi_K Z\|_F^2 = t_K(Z)^\mathsf{T} \Gamma^{-1} t_K(Z). \tag{44}$$

Equivalently, if $S_K(c) = \sum_i c_i K_i$ and $\operatorname{spr}(A) = \lambda_{\max}(A) - \lambda_{\min}(A)$, the spread-pencil identity is

$$m(K) = 2 - \max_{c \neq 0} \frac{\operatorname{spr}(S_K(c))^2}{c^\mathsf{T} \Gamma c}. \tag{45}$$

Indeed, maximizing $\operatorname{tr}(AZ)$ over $Z \in \mathcal{W}$ gives $\operatorname{spr}(A)$, and duality with (44) gives (45).

For symmetric coordinate matrices $A_1, \dots, A_5$, write

$$F_A(a, b) = \sum_{j=1}^{5} \operatorname{tr}\Big(A_j(aa^\mathsf{T} - bb^\mathsf{T})\Big)^2, \qquad (a, b) \text{ orthonormal}, \tag{46}$$

and let $m_A$ and $M_A$ be its minimum and maximum. For an injective map $\Psi$, define

$$c_\Psi = \inf_{R_1 \neq R_2} \frac{\|\Psi(R_1) - \Psi(R_2)\|}{d_\angle(R_1, R_2)}, \quad C_\Psi = \sup_{R_1 \neq R_2} \frac{\|\Psi(R_1) - \Psi(R_2)\|}{d_\angle(R_1, R_2)}, \quad \kappa_\Psi = C_\Psi / c_\Psi$$

when $c_\Psi > 0$.

**Lemma 7.5** (Rotation-angle reduction)**.** *Suppose a map $\Psi : \mathrm{SO}(3) \to \mathbb{R}^5$ and symmetric matrices $A_1, \dots, A_5$ satisfy*

$$\Psi(R(q)) - \Psi(R(r)) = \Big(\operatorname{tr}\Big(A_j(qq^\mathsf{T} - rr^\mathsf{T})\Big)\Big)_{j=1}^{5} \tag{47}$$

*for all unit quaternions $q, r$. If $d_\angle$ is the rotation angle, then*

$$\frac{\sqrt{m_A}}{\pi} d_\angle(R_1, R_2) \le \|\Psi(R_1) - \Psi(R_2)\| \le \frac{\sqrt{M_A}}{2} d_\angle(R_1, R_2). \tag{48}$$

*Proof.* Choose unit quaternions $q, r$ for $R_1, R_2$ and let $\theta \in [0, \pi/2]$ be their line angle. A two-dimensional spectral decomposition gives $qq^\mathsf{T} - rr^\mathsf{T} = \sin\theta\,(aa^\mathsf{T} - bb^\mathsf{T})$ for an orthonormal pair $(a, b)$, while $d_\angle = 2\theta$. Hence $\|\Psi(R(q)) - \Psi(R(r))\|^2 = \sin^2\theta\, F_A(a, b)$, and the bounds follow from $2\theta/\pi \le \sin\theta \le \theta$. Conversely, every direction $aa^\mathsf{T} - bb^\mathsf{T}$ in (42) occurs as the normalized tangent direction of such a pair difference, so the constants $m_A$ and $M_A$ govern the full tangent cone rather than only a subset of it. □

# 8 Conditioning of the positive-projector design

The preceding injectivity construction can now be quantified in the same finite-dimensional model. Let $D$ be the positive diagonal factor in (34). Let $\widehat{K}_{\mathrm{proj}}$ denote the four-plane obtained by whitening the symmetric kernel with the $D$-weighted trace pairing, and let $G_1, \dots, G_5$ be the exact rational representatives of the five trace-projected functionals in those coordinates. Their rational frame Gram is

$$(H_{\mathrm{proj}})_{jk} = \operatorname{tr}(G_j D^{-1} G_k D^{-1}) - \frac{1}{4} \operatorname{tr}(G_j D^{-1}) \operatorname{tr}(G_k D^{-1}). \tag{49}$$

Write $F_{\mathrm{proj}}$ for the realized objective (46) in these coordinates and set

$$s_{\mathrm{lo}} = \frac{8128734403947473536707871850150956025 49}{1050519131516635865783355793260202290053 12}, \tag{50}$$

$$L_0 = \frac{s_{\mathrm{lo}}}{1024} = \frac{8128734403947473536707871850150956025 49}{1075731590673035126562156332298447145014394 88}, \tag{51}$$

$$M_0 = \frac{270988332696903}{281474976710656}. \tag{52}$$

**Theorem 8.1** (Positive-projector kernel and frame-factor floor)**.** *For the positive-projector construction,*

$$m(\widehat{K}_{\mathrm{proj}}) \geq \frac{1}{1024}, \qquad s_{\mathrm{lo}} < \lambda_{\min}(H_{\mathrm{proj}}), \qquad F_{\mathrm{proj}} \geq \lambda_{\min}(H_{\mathrm{proj}}) m(\widehat{K}_{\mathrm{proj}}) \geq L_0.$$

*In particular, the realized five-functional objective has* $\min F_{\mathrm{proj}} \geq L_0 \approx 7.556470846841732 \times 10^{-6}$.

*Proof.* The inequality $m(\widehat{K}_{\mathrm{proj}}) \geq 1/1024$ is proved by exact branch-and-bound in rational arithmetic. The compact coefficient domain in (45) is covered by finitely many boxes. Rational quadratic bounds discard boxes outside the normalized coefficient shell. Vertex-cone bounds close every remaining box in the supplied cover. Vertex eigenvalue bounds use exact integer characteristic polynomials and outward derivative-test bisection. The complete replay supplies all 87 525 closed and 4 626 pruned leaves, with no unresolved box. The supplementary proof guide explains the convexity step and the replay checks the entire subdivision, not merely the earlier one-in-100 sample.

The characteristic polynomial of the rational matrix $H_{\mathrm{proj}}$ and a Sturm root count give $H_{\mathrm{proj}} > 0$ and the strict bracket $s_{\mathrm{lo}} < \lambda_{\min}(H_{\mathrm{proj}})$. The realized five coordinates are a frame analysis of $\Pi_{\widehat{K}_{\mathrm{proj}}^{\perp}} Z$; the Rayleigh quotient for (49) therefore gives

$$F_{\mathrm{proj}} \geq \lambda_{\min}(H_{\mathrm{proj}}) \|\Pi_{\widehat{K}_{\mathrm{proj}}^{\perp}} Z\|_F^2.$$

Taking the minimum over $\mathcal{W}$ and substituting the two rational lower bounds yields (51). □

**Corollary 8.2** (Positive-projector rotation-angle stability)**.** *For the map* $\Phi$ *in* (39) *and rotation-angle distance* $d_\angle$*,*

$$\frac{\sqrt{L_0}}{\pi} d_\angle(R_1, R_2) \leq \|\Phi(R_1) - \Phi(R_2)\| \leq \frac{\sqrt{M_0}}{2} d_\angle(R_1, R_2). \tag{53}$$

*Consequently*

$$c_{\mathrm{proj}} \geq \frac{\sqrt{L_0}}{\pi} \approx 8.750031830575377 \times 10^{-4}, \qquad \kappa_{\mathrm{proj}} \leq \frac{\pi}{2}\sqrt{\frac{M_0}{L_0}} \approx 560.6809131191.$$

*Proof.* A separate Sturm calculation supplies outward eigenvalue brackets for the positive matrices $B_j$. Since $|a^{\mathsf{T}}B_j a - b^{\mathsf{T}}B_j b| \leq \lambda_{\max}(B_j)$, $M_A \leq \sum_{j=1}^{5} \lambda_{\max}(B_j)^2 \leq M_0$. This is independent of the frame-factor lower bound. Apply Lemma 7.5 with $m_A \geq L_0$. The result uses one-sided bounds; it does not identify the unknown true extrema. □

# 9 A separate rational conditioning construction

It is natural to ask whether a different four-plane improves the floor of Theorem 8.1. The supplementary data specify four rational trace-zero symmetric matrices whose span is a four-plane $K^* \subset \mathcal{S}_0$. Unlike $\widehat{K}_{\mathrm{proj}}$, this plane is not obtained from positive rank-two projections; it is an abstract measurement design. Its floor is nevertheless defined by exactly (43).

**Theorem 9.1** (Floor for the rational four-plane)**.** *For the rational four-plane $K^*$ reproduced in the supplement,*

$$m(K^*) \geq \frac{1}{32}.$$

*Proof.* Insert the four rational basis matrices into (45). Exact branch-and-bound, using the same rational interval and vertex-cone procedure as above, proves that the normalized squared spread is at most $63/32$. Equation (45) then gives $m(K^*) \geq 1/32$. The finite cover, consisting of 1 786 closed and 285 pruned leaves, and the engine recomputing its rational bounds are supplied in the supplement. □

The two designs may therefore be compared only at the level of their proved lower-bound values. The floor $1/32$ for $K^*$ is exactly 32 times the positive-projector floor $1/1024$. Likewise,

$$\frac{1/(\pi\sqrt{32})}{\sqrt{L_0}/\pi} = \frac{1}{\sqrt{32L_0}} \approx 64.3081.$$

This is not a ratio of the unknown true minima or true stability constants. The rational four-plane has the stronger proved guarantee, while the positive-projector design retains the constructive tomography interpretation.

Exploratory floating-point optimization gives approximately $m(\widehat{K}_{\mathrm{proj}}) = 0.0010212$ and $m(K^*) = 0.10407$. These are uncertified estimates, not proofs of global optimality. They suggest that the projector floor is comparatively tight while the rational floor is conservative; none of the inequalities above depends on them.

**Corollary 9.2** (An abstract isometric rotation-angle bound)**.** *Let $E = K^{*\perp} \cap \mathcal{S}_0$ and $H = \{x \in \mathbb{R}^6 : \sum_{j=1}^{6} x_j = 0\}$. Both spaces have dimension five, so choose any linear isometry $T : E \to H$ and define $\Psi_6(R(q)) = T\Pi_E(qq^{\mathsf{T}} - I/4)$. This is well defined under $q \sim -q$ and realizes pair differences by*

$$\Psi_6(R(q)) - \Psi_6(R(r)) = T\Big(\Pi_E(qq^{\mathsf{T}} - rr^{\mathsf{T}})\Big) \tag{54}$$

*for all unit quaternions $q, r$. Define $L : \mathbb{R}^6 \to \mathbb{R}^5$ by*

$$L_{ij} = \begin{cases} 1/\sqrt{i(i+1)}, & j \le i, \\ -i/\sqrt{i(i+1)}, & j = i+1, \\ 0, & j > i+1, \end{cases} \qquad 1 \le i \le 5, \quad 1 \le j \le 6. \tag{55}$$

*Then $LL^{\mathsf{T}} = I_5$ and $\|Lx\|_2 = \|x\|_2$ for $x \in H$. For the five-coordinate map $\Phi^* = L\Psi_6$ and rotation-angle distance $d_\angle$,*

$$\|\Phi^*(R_1) - \Phi^*(R_2)\| \ge \frac{1}{\pi\sqrt{32}} d_\angle(R_1, R_2), \qquad c_{\Phi^*} \ge \frac{1}{\pi\sqrt{32}} > 0.0562.$$

*The quaternion dictionary makes these fixed linear coordinates on* SO(3)*; the conclusion concerns this abstract $K^*$ construction, not a positive-projector or device realization.*

*Proof.* The rows of $L$ are orthonormal, and its kernel is the span of $(1, 1, 1, 1, 1, 1)$, proving the asserted isometry on $H$. Hence (54) and Theorem 9.1 make the realized objective equal to $\|\Pi_E Z\|_F^2$ with lower bound $1/32$ on $\mathcal{W}$. Apply Lemma 7.5. □

The distinction between injectivity and stability in abstract phase retrieval, including compact-group formulations, is treated in the primary literature [5, 29, 3].

# 10 Fixed-linear optimality

Wang and Xu define the minimal number $m_{\mathbb{R}}(d)$ of generalized real quadratic measurements required for phase retrieval and prove $m_{\mathbb{R}}(4) = 6$ [28, Theorem 5.1 and the following remark]. The six rank-two projection construction used above originates in Xu's low-rank-recovery and projection phase-retrieval construction [30]; Theorem 7.2 supplies the exact specialization needed here. These generalized quadratic counts are not general tomography counts: minimal qubit tomography and tomography under prior information use different objects and measurement classes [22, 14].

**Corollary 10.1** (Fixed-linear optimality)**.** *By the published theorem $m_{\mathbb{R}}(4) = 6$, five is the minimum number of fixed real-linear functionals on the standard embedded* SO(3) $\subset \mathbb{R}^9$ *that can be globally injective.*

*Proof.* Suppose four fixed linear functionals were injective on SO(3). Composing them with the homogeneous quadratic map $q \mapsto R(q)$ gives four generalized quadratic measurements on $\mathbb{R}^4$. Adjoin the norm measurement $q^{\mathsf{T}}q$. If two nonzero vectors have the same five quadratic values, their norms agree; after normalization, injectivity on SO(3) identifies their unit quaternions up to sign. Hence these five quadratics would perform real phase retrieval in $\mathbb{R}^4$, contradicting $m_{\mathbb{R}}(4) = 6$. □

The value $m_{\mathbb{R}}(4) = 6$ is a published input and is not reproved here. The conclusion concerns fixed real-linear functionals, not adaptive queries, nonlinear encodings, primitive physical readings, or experimental cost. There is also an independent topological obstruction:

the compact space SO(3), homeomorphic to $\mathbb{RP}^3$, does not embed in $\mathbb{R}^4$, although it embeds in $\mathbb{R}^5$ [31, Sec. 4.1]. A continuous injective map from a compact space into Euclidean space is an embedding, so this rules out every continuous four-coordinate encoding, not only fixed linear functionals. The phase-retrieval reduction above gives a second proof within the declared quadratic measurement model.

# 11 Adaptive five-reading implementation comparison

Here *minimal* is relative to the column-first real-linear model below. It does not transfer to POVM tomography, trace-one state estimation, or arbitrary prior information [22, 14].

**Definition 11.1** (Column-first adaptive scheme)**.** For $C = [u\ v\ z] \in \mathrm{SO}(3)$, first read all three entries of $u$. Choose $p \in \arg\max_i |u_i|$, taking the smallest index in a tie. Then read the two components $v_j$ with $j \neq p$, set

$$v_p = -\frac{\sum_{j\neq p} u_j v_j}{u_p}, \tag{56}$$

and complete $z = u \times v$. The resulting matrix can be checked directly for unit columns, orthogonality, and positive orientation.

**Proposition 11.2** (Scoped adaptive sufficiency and minimality)**.** *The column-first scheme reconstructs every $C \in \mathrm{SO}(3)$ from five support-aware readings. After a full first column has been fixed, one additional real-linear reading cannot be injective on the residual circle, whereas the two selected nonpivot entries are injective. Thus five is necessary and sufficient within this declared column-first real-linear scheme.*

*Proof.* A unit vector $u$ has $|u_p| \geq 1/\sqrt{3}$ at the selected maximum-magnitude component. Orthogonality $u^{\mathsf{T}}v = 0$ gives (56); the two nonpivot coordinates therefore determine $v$ uniquely. Exact checks certify $u, v$ as an orthonormal pair, and $u \times v$ chooses the orientation-preserving completion.

Once $u$ is fixed, write the residual column as $v(\phi) = a\cos\phi + b\sin\phi$ in an orthonormal basis of $u^{\perp}$. Any one real-linear reading is $A\cos\phi + B\sin\phi$ and has an exact collision on the circle. By contrast, equality of the two nonpivot components and orthogonality force equality of the pivot component, hence of $v$. □

**Proposition 11.3** (Conditional stability of the adaptive pivot)**.** *Suppose the first column $u$ is known exactly. If the two measured nonpivot coordinates of $v$ have error vector $e \in \mathbb{R}^2$, the reconstruction (56), before any normalization or orthogonality correction, has error*

$$\|\delta v\| \leq \frac{\|e\|}{|u_p|} \leq \sqrt{3}\,\|e\|. \tag{57}$$

*Thus maximum-magnitude pivoting gives a uniform bound for this step without another reading.*

*Proof.* Since $\|u\| = 1$, the selected pivot satisfies $|u_p| \geq 1/\sqrt{3}$. Writing $u_{-p}$ for its two nonpivot coordinates, the pivot error is $\delta v_p = -u_{-p}^{\mathsf{T}} e/u_p$. Cauchy–Schwarz gives

$$\|\delta v\|^2 = \|e\|^2 + \frac{|u_{-p}^{\mathsf{T}} e|^2}{|u_p|^2} \leq \left(1 + \frac{1 - |u_p|^2}{|u_p|^2}\right) \|e\|^2.$$

□

This is not a full noisy-instrument theorem: errors in $u$, pivot selection from noisy first-column data, and subsequent completion are not covered. Choosing the first nonzero component would preserve exact reconstruction but allow an arbitrarily small denominator. The supplement supplies an exact-rational regression of the pivot rule and this distinction; the general bound follows from the proof above, not from those examples.

| Instrument | Count | Mathematical scope |
|---|---|---|
| Antisymmetric $w(C)$ | 3 | Local chart on the positive-cosine branch |
| $w(C)$ plus trace | 4 | Fails globally on every half-turn |
| Column-first scheme | 5 | Global, exact, adaptive; minimal only in its declared model |
| Functionals $\ell_1, \ldots, \ell_5$ | 5 | Global, exact, fixed and nonadaptive; quantitative positive-projector bounds; fixed-linear optimality by $m_{\mathbb{R}}(4) = 6$ |
| Rational $K^*$ coordinates | 5 | Separate abstract design; proved floor $1/32$ and abstract lower constant; not a projector or physical implementation |

Table 1: The adaptive result is an implementation comparison, not an open substitute for the fixed theorem.

## 12 Limitations and related directions

The reconstruction theorems concern exact finite-dimensional data; Proposition 11.3 adds only the stated conditional perturbation bound. The loops are prescribed mathematical paths in $\mathcal{P}$, and their return action is relational in the sense of (21); no physical realization of an arbitrary loop is asserted. Likewise, the five fixed coordinates are synthesized linear functionals, not five primitive experimental shots. The fixed-linear lower bound uses the published identity $m_{\mathbb{R}}(4) = 6$, and the adaptive minimality statement applies only to the column-first model of Definition 11.1.

The conditioning estimates concern the declared exact coordinate maps. They do not establish stochastic-noise robustness, sample complexity, device performance, or optimality among all four-planes. In particular, the lower angle constant for $K^*$ applies to the abstract realization (54), not to an unspecified device. Robust synchronization under corruption is a natural comparison direction [23, 6, 17, 12], but incorporating noise into the Green-return reconstruction problem remains open.

# 13 Conclusion

The Green pairing determines the affine-invariant geometry of positive Hermitian $2 \times 2$ instruments. Its determinant-one shape leaf is hyperbolic with the exact normalization $K = -1/2$, and its exact transport turns closed instrument loops into relational SO(3) return actions. Three readings are locally sufficient, the natural fourth trace reading does not cross the half-turn wall, and five fixed synthesized functionals are globally injective. Their proof is constructive: six exact rank-two projections, exact whitening, an exact quaternion–rotation dictionary, and a complete projective real-kernel exclusion. Fixed-linear optimality remains explicitly dependent on the published value $m_{\mathbb{R}}(4) = 6$. The adaptive five-reading scheme remains an elementary alternative within its narrower information model.

The positive-projector map has $m(\widehat{K}_{\mathrm{proj}}) \geq 1/1024$, $\min F_{\mathrm{proj}} \geq L_0$, the two-sided estimate (53), and $\kappa_{\mathrm{proj}} \leq 560.6809131191\ldots$. The separate rational plane $K^*$ has $m(K^*) \geq 1/32$ and, for the constructed isometric coordinates, a lower angle constant exceeding 0.0562. The factor 32 in the floor bounds and the factor $64.3081\ldots$ in the corresponding proved lower constants compare guarantees, not unknown true optima. Determining the best four-plane, and understanding which well-conditioned abstract designs admit natural positive or physical realizations, are the principal open problems suggested by this analysis.


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